

My Project: LOCKD

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Contents

Introduction.....	3
Software Development Lifecycle	3
Sprint Backlog.....	5
Sprint 1.....	5
Sprint 2.....	5
Sprint 3.....	6
Sprint 4.....	6
Sprint 5.....	6
Sprint 6.....	7
Sprint 7.....	7
Sprint 8.....	8
Design Document	9
Project Vision	9
Background.....	9
UML diagrams	10
Sitemap.....	10
Wireframes.....	11
Log In Sequence Diagram	13
Algorithm Design	13
Salt/Key Generation	13
Encryption and Decryption	13
Hashing.....	14
Noted Issues and Constraints.....	15
LESP Issues.....	15
GitHub Repository Link	15
Appendix	16
Testing Table.....	16
References.....	17

Introduction

This report presents my project plan for the coursework of the COMP1004 module. The project mainly focuses on the Software Development Lifecycle (SDLC) and personal time management. My chosen project is a secure notes website which I have called LOCKD. The website's main focus will be on security, as that is my chosen pathway within the computer science degree. Within this report I will first discuss the SDLC and explain how I have applied this to my project. I will then provide a design document underlining my design process, project vision and other relevant information. Following this I will then explain any issues and constraints I encountered and finally I will link my Github repository.

Software Development Lifecycle

The software development lifecycle defines the steps in software development and the order they're executed. It is used to make software development more efficient and there are many models. In this project the agile philosophy model called scrum has been used, where the project is split into sections called sprints and every set period (in this case 2 weeks) a functional piece of the project should be delivered (Tiwari, 2025). Scrum usually includes a scrum team, developers, product owner and a scrum master who all work on the project, fulfilling different tasks (Schwaber & Sutherland, 2020); however, as this is an individual project I shall be taking on all these roles, albeit with the specific tasks slightly modified.

The planning phase was first, where different website ideas were considered, with relevance to my computer science pathway of cyber security a priority. After some time, the final idea of a secure note app was decided, as it has elements of cyber security but is not too complex for the scope of this module. Research into websites like this was carried out, to establish what the current websites on the market are like and what could be done to make mine stand out. Standard notes (2026) was one of them and some design elements from it was noted down as inspiration for the design phase.

Having researched and decided on an idea for the project, with elements from other sites noted, requirement analysis started with both functional and non-functional requirements considered. The main 3 requirements were encrypted files; salted and hashed passwords; and user created passwords must be secure. These requirements were made to ensure the application is as secure as possible.

Once all the requirements were created, user personas were made, such as user and developer. Then user stories, which are a 'simple description of a software feature written from the end user's perspective' (Talreja, 2025), were created with requirements mapped to them. These were then used to make the Product Backlog (fig. 1), which is a list of work needed to be done that guides developers on what to deliver (Maria, 2025).

The Sprint Backlog was then created, with a rough idea of a sprint plan laid out. User stories and tasks were mapped to each sprint, with a deliverable set out explaining what was expected at the end of each sprint. A Gantt chart was also created, which is a visual way of displaying what is expected to be completed and when, using a horizontal bar chart (LaPrad, 2026).

This concluded the sprint planning, and then the 2 weeklong sprints commenced. In the first sprint I set up my infrastructure and started creating some Unified Modelling Language (UML) diagrams, which helped design the page layouts and system design. At the end of each sprint, I

went over what I had set out to do and if I had not completed all tasks then that was pushed to the next sprint. I also reviewed, what I had planned for the next sprint and modified it if needed. My sprints are all shown in the Sprint Backlog section later.

The screenshot shows a web application interface for a 'Product Backlog'. At the top, there is a header bar with the text 'Product Backlog • Saved' and a search bar labeled 'Search'. Below the header, there is a navigation menu with links: 'Layout', 'References', 'Mailings', 'Review', 'View', 'Help', and 'Turnitin'. The main content area displays a list of tasks for a 'Secure Notes Application'. The tasks are organized into sections: 'Website Infrastructure Setup', 'Layout Design', 'Architecture', 'Login Page', and 'Home Page'. Each section contains a user story and a list of tasks.

Product Backlog • Saved

Search

Layout References Mailings Review View Help Turnitin

Secure Notes Application

Website Infrastructure Setup

- User story: As a developer, I want to set up the infrastructure so I can start building the website
 - o Tasks:
 - Set up repository
 - Download all needed extensions

Layout Design

- User story: As a user, I want the website to be easy to navigate
 - o Tasks:
 - Create sitemap
 - Create base wireframe
 - Create individual 'page' wireframes

Architecture

- User story: As a developer, I want the website to be easy to program and maintain
 - o Tasks:
 - Create the package diagram
 - Create class diagrams
 - Create the sequence diagrams

Login Page

- User story: As a user, I want my notes to be kept secure behind a login page
 - o Tasks:
 - Design and implement a login page, that after a successful login, opens to the home page
 - Securely hash and store the password

Home Page

- User story: As a user, I want to be able to easily see all my notes and access them
 - o Tasks:
 - Design and implement layout
 - Add previews of notes
 - Notes [previews](#) link to the whole note

Figure 1: Product Backlog

Sprint Backlog

This section shows my sprint backlog, which is how I planned all my sprint. It includes the user stories, tasks and deliverables of each sprint; as well as a guide as to how well each sprint was completed.

Sprint 1

26/11/2025 – 9/12/2025

User Stories

- As a developer, I want to set up the infrastructure so I can start building the website
- As a user, I want the website to be easy to navigate
- As a user, I want all my notes to be encrypted to ensure that they are secure

Task	RAG
Set up repository	
Download extensions and required software	
Create Sitemap (fig. 2)	
Create wireframes (figs. 3 to 9)	

Deliverables

The software is set up, so when the design is ready, I can start programming. The layout of the website is designed so program architecture can now be designed.

Sprint 2

9/12/2025 – 23/12/2025

User Stories

- As a developer, I want the website to be easy to program and maintain

Task	RAG
Research Best Encryption Methods	
Research best security practices	
Design the security for the website	

Deliverables

The architecture of the code is designed, so when programming starts it is easy to know what to code and the security has been researched and designed.

Issues

Research for the security led to me realising that the initial idea of using End-To-End encryption to stand out from other notes applications was too complicated and out of scope for this particular project. So, the security for the website was initially designed here using node.js crypto module, but no end-to-end encryption, however this was changed later as it became clear this too was out of scope.

Sprint 3

1/1/2026 – 15/1/2026

User Stories

- As a developer, I want to complete the mid project review

Task	RAG
Create the slides	Green
Record the video	Green
Write my professional development review	Green

Deliverables

The slides and video explaining the scope of my project and work so far will be done and my professional development review will be submitted.

Sprint 4

15/1/2026 – 29/1/2026

4/2/2026 – 18/2/2026

User Stories

- As a user, I want my notes to be kept secure behind a login page
- User story: As a user, I want to be able to modify the design of the website to my liking

Task	RAG
Design and implement a login page, that after a successful login, opens to the home page	Green
Securely hash and store the password	Red
Implement the layout of the page (fig. 4)	Green
Add settings to change the design (theme, font size, language) (fig. 9)	Green

Deliverables

The login page will be complete, with the username and password stored securely. The settings button will also be complete with the drop-down working and buttons to change the font size, theme and language.

Issues

I made the frontend of the log in page but then hit issues with the encryption and storing the password securely. I wasn't getting anywhere with it, so I ended up sorting out the settings button instead. This means part of a later planned sprint from my Gantt chart is done, so I am not too far behind because of this issue. I plan on fixing the encryption issues in the next sprint.

Sprint 5

18/2/2026 – 4/3/2026

User Stories

- As a user, I want my notes to be kept secure behind a login page
- As a user, I want to be able to easily see all my notes and access them

Task	RAG
------	-----

Securely hash and store the password	
Implement layout of home page (fig. 6)	
Make button to link to whole note	
Make button that will create a new note	

Deliverables

Username and passwords for the log in will be hashed and stored securely. The home page will be finished, with all buttons working to an extent, as the note page will not be done yet.

Sprint 6

4/3/2026 – 18/3/2026

User Stories

- As a user, I want to be able to edit my notes and clear them

Task	RAG
Design and implement note layout (fig. 8)	
Add buttons (delete, clear and save)	
Add ways to change design of note (font, font colour, font size etc)	
Add way to pin certain notes	

Deliverables

Home page will be able to link to the note page and note page will be fully functional, with files saving in browser local storage.

Issues

There were many issues with implementing ways to change the design of the note. I was using the text area element, which does not allow the text in it to be altered in anyway. So, I tried changing to a content editable element, however, to retrieve the selection that the user highlighted and then change it required libraries which are not allowed to be used. So, after a lot of trial and error, I decided to not implement these ideas as they were just functional requirements. My focus on these elements also meant I neglected the other tasks.

Sprint 7

18/3/2026 – 1/4/2026

User Stories

- As a user, I want to be able to edit my notes and clear them
- As a user, I want to be able to easily see all my notes and access them
- As a user, I want all my notes to be encrypted to ensure that they are secure

Task	RAG
Add buttons (delete, clear)	
Add way to unpin note	
Button for user to log out	
Implement preview page (fig. 7)	
Research best encryption methods	
Research best security practices	
Encrypt each file	
Decrypt each note when selected	

Decrypt part of each note as preview	
--------------------------------------	--

Deliverables

Home page will link to preview and then that will link to full note view where users will be able to unpin, clear and delete notes. Users will also be able to log out. All notes will be encrypted when stored in local storage, then when retrieved they'll be unencrypted.

Sprint 8

1/4/2026 – 15/4/2026

User Stories

- As a developer, I want all notes stored in their own file

Task	RAG
Store all notes in their own JSON file	
JSON Files can be uploaded	
Give users unique IDs	

Deliverables

All notes will be stored in a JSON file on users' computer and if deleted from browser local storage, can be uploaded back to the website. User's will also have their own unique ID so only they can access their files.

Design Document

This section explains and shows how my project was designed and the reasons for the way it was done.

Project Vision

I will develop a notes application that will use the necessary encryption to ensure all notes will be stored securely and will be inaccessible by unauthorised users. It will give users a place to store sensitive data, such as passwords for other applications, or even data that they just want to keep a secret from others, such as present ideas.

There are other secure notes apps, such as standard notes (2026), which allows users to customise the app in many ways, that are like my project. I will take some inspiration from them by letting users customise the app, but they will have less options like only changing the theme and font size of the app, so it is easier to navigate.

Each user's username and password will be stored securely, so that all users can feel safe in the knowledge that, when they are creating and editing notes, their data is safe from any unauthorised access. I will be storing the users' passwords using a hashing algorithm of my own design, which is based on MD5. MD5 is not the most secure as collisions are possible (CipherUtility, 2025), however a more secure algorithm is difficult to implement and out of my current skill set.

Users will also be able to, once logged in, see the titles of all their notes, but not the actual data stored (fig. 3). Once they've clicked on the title, they will get a few sentence 'preview' of the note (fig. 7) and then the option to edit and see the whole note (fig. 8). This ensures that only necessary data is shown, reducing the risk of shoulder surfers seeing data that the user doesn't need to access at that point in time.

Background

LOCKD is a secure notes website with a focus on keeping all notes secure from unauthorised access. As cyber-attacks become more common with "204 'nationally significant' cyber-attacks against the UK" between September 2024 to August 2025, a rise from the 89 the previous year (NCSC, 2025), people are wanting more secure ways to store data. The majority of note apps or websites skip end-to-end encryption and don't allow you to easily store data locally (Shah, 2024) such as Google Keep (2026) and UpNote (2026). So, LOCKD will implement both these ideas, so users feel safe in the knowledge that their data is being stored securely.

Accessibility is also a key part of my website, as it makes it available to more people, but it is also a legal requirement (Gov.uk, 2024). To ensure my website is accessible I have included options in settings (fig. 9) to make the font larger so people with poor eyesight are able to read it. I have also included, in the HTML, the lang tag, so a screen reader knows the language it is reading. Finally, I have made sure all elements of my website have relevant and easy to understand IDs, so again screen readers can identify the button or input box that has been selected.

UML diagrams

This section shows all the diagrams I have created to help the implementation of my website.

Sitemap

This section shows the sitemap and how I expect my single page application 'pages' to be linked together.

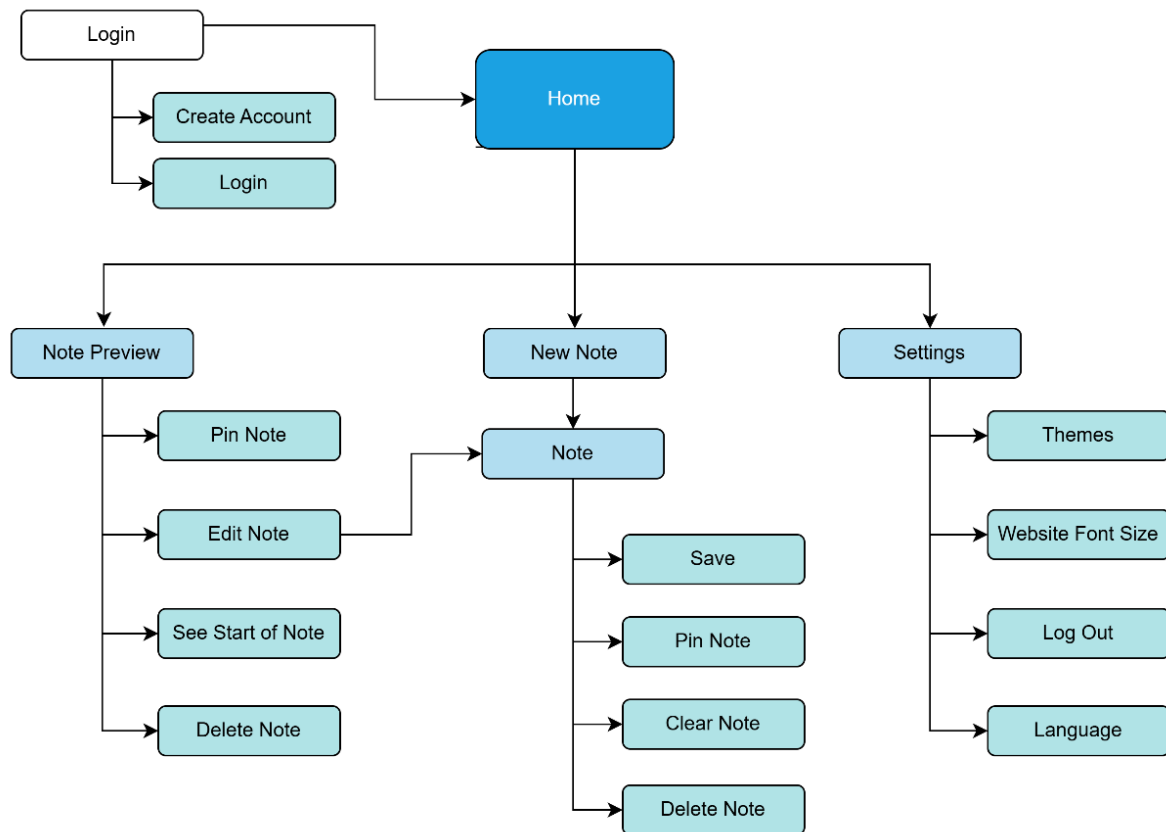


Figure 2: UML diagram illustrating the structure of the SPA

Wireframes

This section provides my design of how the website should look with the use of wireframes.

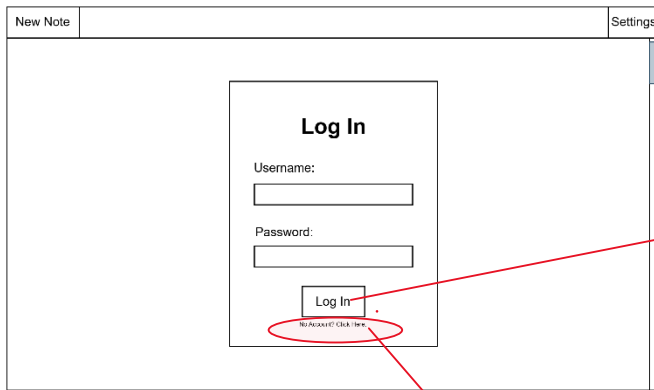


Figure 4: UML diagram illustrating the Log In page



Figure 3: UML diagram illustrating the Home Page

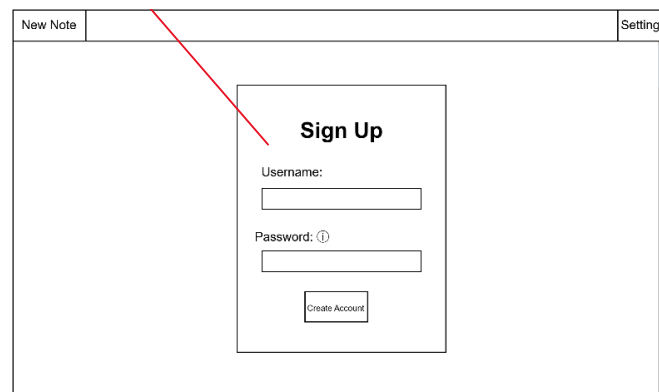


Figure 5: UML diagram illustrating the Sign Up Page

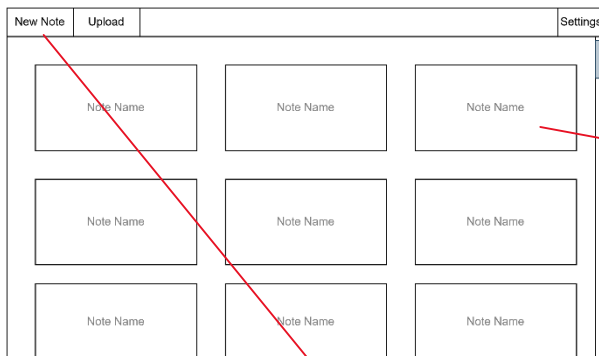


Figure 6: UML diagram illustrating the Home Page

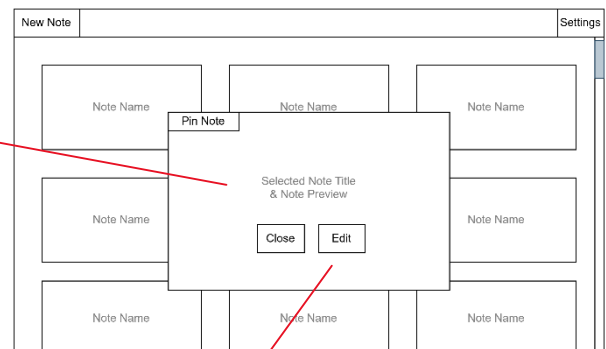


Figure 7: UML diagram illustrating the Preview Page

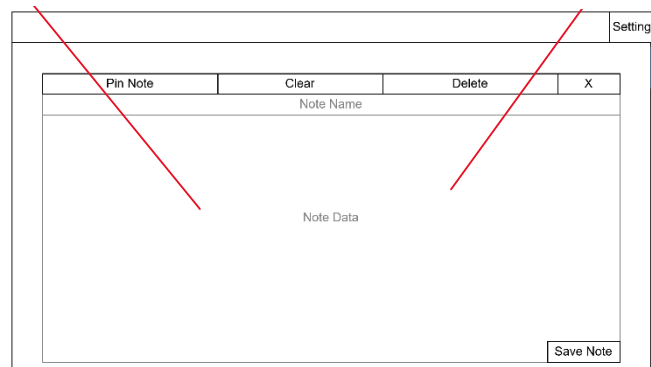


Figure 8: UML diagram illustrating the Note Page

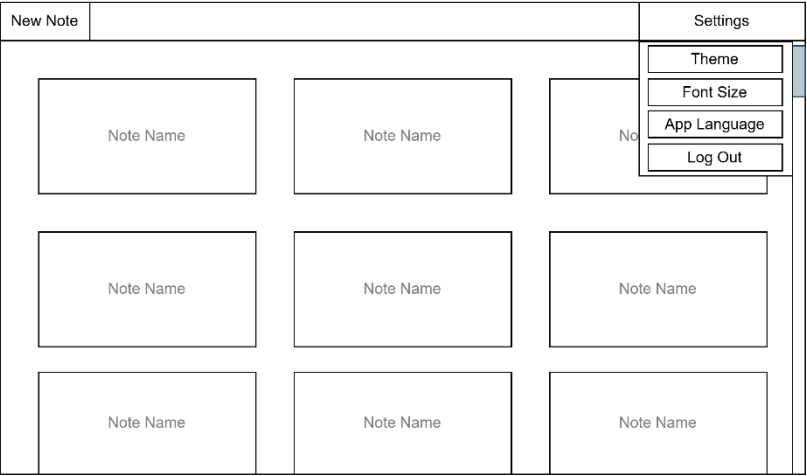


Figure 9: UML diagram illustrating the Settings Pop Up

Log In Sequence Diagram

This diagram shows how a user, once they have input their username and password and clicked the log in button, will be validated against the stored data.

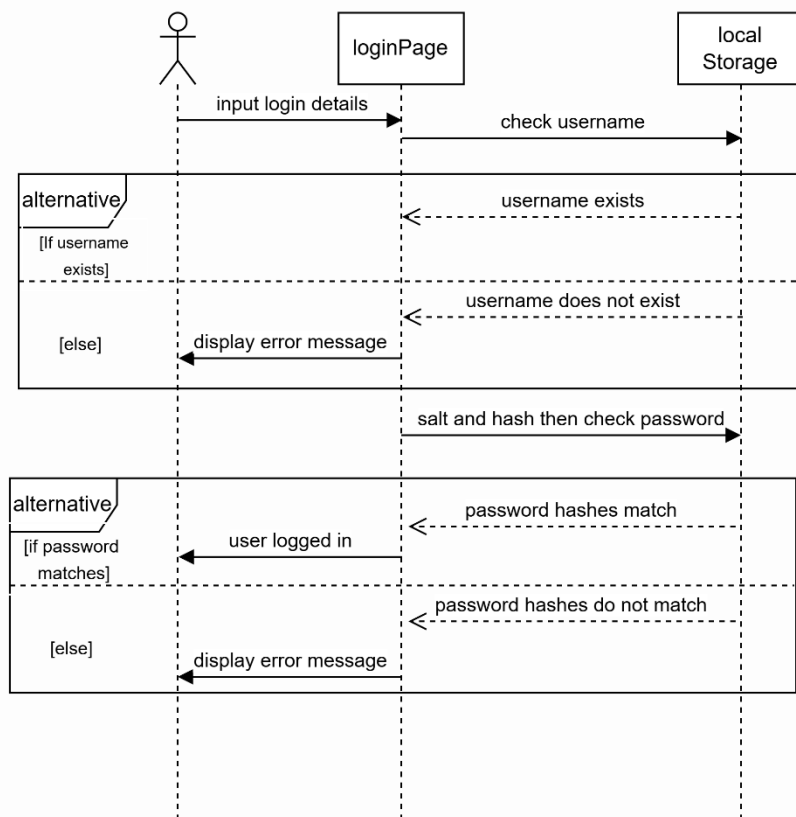


Figure 10: UML diagram illustrating the sequence of logging in

Algorithm Design

This section shows the design of all my algorithms to do with security.

Salt/Key Generation

This section shows the design of my key generation algorithm. The result that is generated is used as the salt for salting and hashing, as well as the key in the encryption algorithm.

1. Generate a random number between 33 and 126
2. Convert the random value to its corresponding character
3. Concatenate to salt variable
4. Repeat until the salt variable is 16 characters long

Encryption and Decryption

This section shows the design of my algorithm that both encrypts and decrypts given values.

1. Convert given string to binary
2. Convert given key to binary
3. XOR 1 bit from the string and 1 bit from the key
4. Concatenate the result to the encrypted string variable
5. Repeat steps 3 and 4 for the length of the binary string

6. Convert the encrypted string from binary to a string

Hashing

This section shows how the hashing algorithm works. It is based on the algorithm MD5, which is a cryptographic hash function (Kumar, 2025); however, it is simplified to make it easier to implement.

1. Concatenate the given salt to the given password
2. Convert the salted password to binary
3. Pad the salted password with a 1 and then 0s until its length is a multiple of 512
4. Set variables A, B, C and D with set hexadecimal values
5. Split the binary salted password into 32-bit chunks and store in an array
6. Repeat the next steps 4 times
7. Loop each block of the binary password
8. Swap variable A with D, C with B and D with C
9. For each iteration, calculate a value with the corresponding calculation
 - a. $(B \text{ AND } C) \text{ OR } (\text{NOT } B \text{ AND } D)$
 - b. $(B \text{ AND } D) \text{ OR } (C \text{ AND } \text{NOT } D)$
 - c. $B \text{ XOR } C \text{ XOR } D$
 - d. $C \text{ XOR } (B \text{ OR } \text{NOT } D)$
10. Add the result, A and the current password block together
11. Assign Sum to B, A back to A, C back to C and D back to D.

Noted Issues and Constraints

One issue I encountered during this project was due to the constraints given by the allowed technologies list, as well as my limited understanding of node.js and its crypto module. I originally wanted to use the crypto module for all my cyber security issues, like encryption, salting and hashing, however after a couple weeks of research while I delayed implementing the security and instead focused on implementing the settings alongside my research, I concluded that I would not be able to use crypto. I was unable to figure out how to use it alongside html as it would not work browser side and using a server is out of the scope of this project. I found the browserify library (ljharb, 2024) which was the only way I could think of fixing it but realised, I could not use it due to it not being on the allowed technologies list. In the end I decided to implement my security with my own encryption and hashing algorithms which means the project is not as secure as I'd like, but at least it has some security implemented.

Another issue I encountered was time management. Alongside this project I also had 2 other pieces of coursework due within a week of this one's due date. I had to try and balance my focus of all 3 coursework, however I preferred this coursework as it includes programming and I enjoy that, so I tended to focus on this more than the other ones. As a consequence of this, I fell behind on those two coursework and to fix this I had to set some days aside where I ignored this coursework to focus on the others. I also worked for 2 weeks over the easter holidays, which meant my sprint during that time was very dialled back and I could not complete much, so the next sprints ended up having more to do.

LESP Issues

I also had to consider any legal, social and ethical issues. Customer data, including passwords, usernames and note data, all have to be kept secure according to (General Data Protection Regulation, 2016). So, to ensure this, I implemented salting and hashing on passwords and encrypted all notes stored. This meant that even if an attacker got access to the data, they would struggle to be able to read and understand it. Another legal issue I had to consider was accessibility, which I explained in the background section earlier.

One of the ethical issues I had to consider was usability. I had to make sure the website was intuitive and had consistent design so my users would have a pleasant experience and would be more likely to come back to the website. To do this I made sure all buttons had clear names; appropriate alerts would appear if errors occurred, ensuring users understand what went wrong; and finally having pop ups appear to make sure users know what they're about to do (like delete a file).

GitHub Repository Link

<https://github.com/hedwignu/HeidiGinnaw-COMP1004>

Note: I worked on a branch called test for a couple of months and only merged it to master recently, so a lot of my commits are on test.

Appendix

Testing Table

This section shows my testing table, which shows basic test of individual elements and their expected outcome, alongside their actual outcome.

Test NO.	Test	Expected Outcome	Actual Outcome
1	Setting button clicked	Opens drop down	Opens drop down
2	Theme button clicked when on light mode	Changes theme to dark	Changes theme to dark
3	Theme button clicked when on dark mode	Changes theme to light	Changes theme to light
4	'+' font size button clicked when on medium font	Font increases to large	Font increases to large
5	'+' font size button clicked when on large font	Nothing happens	Nothing happens
6	'-' font size button clicked when on large font	Font decreases to medium	Font decreases to medium
7	Language button clicked when in English	Website changes to Swedish	Website changes to Swedish
8	Correct username and Password input and log in button clicked	Logs user in and opens home page	Logs user in and opens home page
9	Incorrect username and password input and log in button clicked	Error message pops up	Error message pops up
10	Plaintext file saved	Stored in an encrypted format	Stored in an encrypted format
11	Encrypted note selected	Opens to preview and is decrypted correctly	Opens to preview and is decrypted correctly
12	Continue button on preview is selected	Opens to full note page with fully decrypted text	Opens to full note page with fully decrypted text
13	Upload button clicked	File selector opens, with only JSON files allowed	File selector opens, with only JSON files allowed
14	File selected from file selector	File added to note homepage with correct data	File added to note homepage with correct data
15	Delete button selected	Warning message appears	Warning message appears
16	Delete confirm button selected	File is deleted from browser local storage	File is deleted from browser local storage
17	Pin button is clicked when note is not pinned	Note is added to the pin list and is displayed at the top	Note is added to the pin list and is displayed at the top
18	Pin button is clicked when note is pinned	Note is removed from the pin list	Note is removed from the pin list

19	Log out button is clicked	User is logged out and sent to log in page	User is logged out and sent to log in page
20	'No account? Click here.' Button clicked	User is sent to the sign up page	User is sent to the sign up page
21	Account is created	Password is salted and hashed and stored in local storage	Password is salted and hashed and stored in local storage

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